

VENTSPILS, Latvia

WMO: 263140

Lat: 57.3956N

Lon: 21.5372E

Elev: 10

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 4.6        | 11.2 | -0.5                          | 5.3 | 6.5  | 5.2 | 7.2 | 12.9 | 29.5                  | 35.8 | 25.8 | 35.6 | 5.8                   | 120  | 0.627 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 7             | 13.5                      | 80.2            | 67.1 | 75.6 | 65.2 | 72.8 | 64.1 | 69.2                | 76.4 | 67.2 | 73.5 | 65.4 | 71.1 | 8.6                     | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 66.5                            | 97.5 | 72.5 | 64.6 | 91.2 | 70.4 | 62.9 | 85.7 | 68.6 | 33.3          | 76.6 | 31.6 | 73.2 | 30.2 | 71.0 | 77.5              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 25.5              | 22.3 | 19.7 | -0.2                       | 85.3 | 8.8                | 5.5 | -6.6                                               | 89.2 | -11.7      | 92.5 | -16.7      | 95.6 | -23.1      | 99.6 |
|                   |      |      | -0.8                       | 72.0 | 8.6                | 3.3 | -7.0                                               | 74.4 | -12.1      | 76.3 | -16.9      | 78.1 | -23.2      | 80.5 |

Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 45.5   | 29.8  | 29.0  | 34.3  | 42.5  | 50.3  | 57.3  | 62.8  | 62.6  | 55.7  | 47.1  | 39.6  | 34.2  |
|                                                                        | DBStd       | 13.52  | 8.79  | 9.11  | 5.86  | 6.52  | 6.98  | 5.37  | 4.97  | 4.55  | 4.91  | 5.84  | 6.54  | 8.01  |
|                                                                        | HDD50       | 2960   | 626   | 587   | 488   | 242   | 77    | 4     | 0     | 0     | 10    | 123   | 312   | 491   |
|                                                                        | HDD65       | 7184   | 1091  | 1007  | 953   | 675   | 459   | 242   | 104   | 102   | 280   | 554   | 761   | 956   |
|                                                                        | CDD50       | 1331   | 0     | 0     | 0     | 18    | 87    | 221   | 397   | 391   | 182   | 34    | 1     | 0     |
|                                                                        | CDD65       | 80     | 0     | 0     | 0     | 0     | 4     | 10    | 36    | 29    | 1     | 0     | 0     | 0     |
|                                                                        | CDH74       | 562    | 0     | 0     | 0     | 1     | 50    | 91    | 243   | 171   | 6     | 0     | 0     | 0     |
|                                                                        | CDH80       | 119    | 0     | 0     | 0     | 0     | 7     | 20    | 55    | 36    | 1     | 0     | 0     | 0     |
|                                                                        | WSAvg       | 9.1    | 10.5  | 9.5   | 9.7   | 9.0   | 8.3   | 8.5   | 7.8   | 7.7   | 8.8   | 9.5   | 9.6   | 10.6  |
|                                                                        | PrecAvg     | 25.6   | 1.8   | 1.1   | 1.3   | 1.3   | 1.3   | 1.9   | 2.7   | 2.6   | 3.0   | 2.8   | 3.3   | 2.6   |
| (15) Precipitation                                                     | PrecMax     | 39.6   | 4.1   | 2.8   | 2.6   | 2.8   | 2.8   | 4.8   | 9.6   | 8.0   | 6.6   | 7.1   | 5.2   | 5.6   |
|                                                                        | PrecMin     | 17.0   | 0.2   | 0.2   | 0.0   | 0.1   | 0.2   | 0.0   | 0.2   | 0.0   | 0.6   | 0.6   | 0.7   | 0.0   |
|                                                                        | PrecStd     | 4.6    | 1.0   | 0.7   | 0.7   | 0.7   | 0.6   | 1.1   | 1.8   | 1.9   | 1.4   | 1.6   | 1.0   | 1.2   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 45.0  | 44.5  | 56.8  | 69.6  | 80.1  | 82.7  | 85.9  | 84.2  | 73.4  | 64.0  | 52.5  | 48.4  |
|                                                                        |             | MCWB   | 44.1  | 41.7  | 46.7  | 53.4  | 63.5  | 66.3  | 69.4  | 70.3  | 61.7  | 56.5  | 49.8  | 47.3  |
|                                                                        | 2%          | DB     | 43.1  | 41.4  | 47.7  | 62.5  | 73.2  | 75.5  | 80.4  | 78.5  | 69.4  | 59.3  | 51.0  | 46.6  |
|                                                                        |             | MCWB   | 41.9  | 39.8  | 42.7  | 52.1  | 61.2  | 64.0  | 68.1  | 66.9  | 61.4  | 55.5  | 49.0  | 45.2  |
|                                                                        | 5%          | DB     | 41.3  | 40.2  | 43.1  | 57.6  | 67.5  | 71.1  | 75.6  | 74.8  | 66.3  | 57.3  | 49.8  | 45.0  |
|                                                                        |             | MCWB   | 40.0  | 38.8  | 40.2  | 48.7  | 58.3  | 61.5  | 66.4  | 65.2  | 60.0  | 54.1  | 48.0  | 43.7  |
|                                                                        | 10%         | DB     | 39.5  | 38.9  | 41.0  | 53.4  | 62.3  | 67.0  | 72.1  | 71.5  | 64.2  | 55.5  | 48.3  | 43.6  |
|                                                                        |             | MCWB   | 38.2  | 37.7  | 38.8  | 46.5  | 55.0  | 59.9  | 65.3  | 64.1  | 59.2  | 52.9  | 46.7  | 42.2  |
|                                                                        | 0.4%        | WB     | 44.0  | 42.1  | 47.3  | 56.0  | 65.7  | 69.2  | 72.3  | 72.1  | 64.6  | 58.2  | 51.0  | 47.4  |
|                                                                        |             | MCDB   | 44.3  | 43.2  | 53.7  | 64.8  | 75.9  | 78.4  | 80.3  | 79.0  | 70.6  | 61.5  | 51.9  | 47.7  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 42.0  | 40.1  | 42.9  | 52.7  | 62.4  | 65.4  | 69.6  | 69.0  | 62.6  | 56.3  | 49.4  | 45.4  |
|                                                                        |             | MCDB   | 42.6  | 40.9  | 45.9  | 60.8  | 70.3  | 73.2  | 76.8  | 75.7  | 66.7  | 59.0  | 50.5  | 46.1  |
|                                                                        | 5%          | WB     | 40.1  | 38.8  | 40.6  | 49.7  | 58.9  | 62.7  | 67.7  | 66.8  | 61.1  | 54.9  | 48.1  | 43.8  |
|                                                                        |             | MCDB   | 40.9  | 39.7  | 42.5  | 56.5  | 66.7  | 69.1  | 74.3  | 72.4  | 65.1  | 57.2  | 49.2  | 44.8  |
|                                                                        | 10%         | WB     | 38.3  | 37.5  | 38.9  | 47.1  | 55.3  | 60.5  | 65.8  | 65.2  | 59.6  | 53.4  | 46.7  | 42.1  |
|                                                                        |             | MCDB   | 39.3  | 38.5  | 40.3  | 52.2  | 61.1  | 65.7  | 71.5  | 70.1  | 63.4  | 55.6  | 48.0  | 43.2  |
|                                                                        | 5% DB       | MDBR   | 7.3   | 8.5   | 9.4   | 13.3  | 14.4  | 13.4  | 13.5  | 14.4  | 12.8  | 10.2  | 7.0   | 6.7   |
|                                                                        |             | MCDBR  | 5.7   | 6.5   | 13.6  | 21.9  | 24.4  | 22.8  | 21.9  | 21.7  | 18.8  | 13.7  | 7.4   | 5.5   |
|                                                                        |             | MCWBR  | 5.8   | 5.8   | 9.7   | 12.8  | 13.4  | 12.9  | 11.9  | 11.7  | 12.4  | 10.4  | 7.0   | 5.7   |
|                                                                        |             | MCWBR  | 5.9   | 6.3   | 12.1  | 19.7  | 22.5  | 19.4  | 18.8  | 17.7  | 15.7  | 11.6  | 6.9   | 5.4   |
| (35) Mean Daily Temperature Range                                      | 5% WB       | MCWBR  | 6.0   | 6.0   | 9.2   | 12.6  | 12.9  | 12.0  | 11.0  | 10.9  | 11.2  | 9.5   | 7.1   | 5.7   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.300 | 0.325 | 0.347 | 0.380 | 0.374 | 0.371 | 0.402 | 0.395 | 0.364 | 0.347 | 0.333 | 0.296 |
|                                                                        | taud        |        | 2.225 | 2.269 | 2.314 | 2.314 | 2.398 | 2.446 | 2.381 | 2.392 | 2.456 | 2.403 | 2.259 | 2.184 |
|                                                                        | Ebn at Noon |        | 176   | 225   | 254   | 264   | 273   | 276   | 263   | 257   | 248   | 217   | 160   | 139   |
|                                                                        | Edh at Noon |        | 16    | 24    | 30    | 35    | 34    | 33    | 35    | 32    | 26    | 21    | 16    | 13    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 105   | 297   | 763   | 1351  | 1836  | 1958  | 1818  | 1441  | 941   | 433   | 137   | 70    |
|                                                                        | RadStd      |        | 18    | 44    | 89    | 153   | 148   | 93    | 158   | 117   | 108   | 55    | 25    | 10    |

Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (28 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page